

Week 7 - Analytics Testing & Refinement

Data Analytics Track | Testing, KPI Validation & Dashboard Refinement

PROJECT OBJECTIVE

Test whether the Week 6 analytical outputs are accurate, consistent and useful for decision-making; refine the dashboard and recommendations where appropriate; and identify the analytical hand-off that would be most useful to Data Science before Week 8.

AnalystLab Africa Experience Lab • Week 7

1. Week 6 → Week 7 Transition

Week 6 moved the HealthConnect Data Analytics work from broad exploratory analysis into a focused no-show risk analysis. The main output was a Power BI decision-support page containing four advanced visuals, a validated overall no-show KPI, a Key Takeaway and Business Insight/Action panels.

Transition item	Week 7 position
Main Week 6 output	Power BI Advanced Analytics & Decision Support dashboard/report.
Most important result	Overall no-show rate of 48.46%: 2,423 no-shows out of 5,000 appointments.
Strongest observed indicators	Previous no-show history and longer booking lead times.
Main limitation/dependency	Observed relationships are not causal; confounding and predictive performance were not fully tested.
Component requiring testing	KPI, four advanced visuals, dashboard interpretation and associated recommendations.
Relevant track(s)	Data Analytics; Data Science is the intended analytical hand-off, but no collaborator output was received.
Week 7 intention	Reconcile KPI calculations, validate documented segment findings, test dashboard consistency/usability, refine interpretation and define Data Science validation requirements.

2. Testing Approach

TESTING SCOPE NOTE: The supplied evidence includes the Week 6 report, dashboard screenshot and PBIX file, but not the original CSV dataset. Therefore this report distinguishes documented/dashboard-level validation from raw-data tests that remain outstanding.

Testing was structured around the most decision-relevant Week 6 outputs rather than repeating the full Week 5 exploratory analysis. The approach uses reconciliation of documented calculations, consistency checks across the Week 6 visuals, review of segment logic, and a refinement review of the dashboard's business interpretation.

Test area	Method	Purpose
Overall KPI	Recalculate no-show rate from documented numerator and denominator.	Confirm the headline KPI is mathematically consistent.
Previous no-shows	Compare documented rates across the main interpretable groups.	Check whether the increasing pattern remains internally consistent.
Booking lead time	Review rates across the five documented lead-time bands.	Check monotonicity and the size of the observed difference.
Reminder channel	Compare the four documented channel rates.	Confirm ordering and ensure wording does not imply causality.
Appointment type	Review the relative closeness of the four categories.	Check whether appointment type is being over-interpreted.

Dashboard refinement	Review whether visuals, KPI and action panels communicate validated findings clearly.	Reduce ambiguity and keep the page decision-oriented.
Cross-track readiness	Translate findings into explicit questions/data requirements for Data Science.	Create a usable hand-off without claiming an uncompleted collaboration.

3. KPI Validation

The core Week 6 KPI was rechecked against the supplied Week 6 Advanced Analytics report and dashboard evidence. The documented calculation is $2,423 \text{ no-shows} \div 5,000 \text{ appointments} \times 100 = 48.46\%$.

KPI	Expected / documented	Validation result	Status
Total appointments	5,000	Documented Week 6 analytical population.	PASS
No-shows	2,423	Documented Week 6 numerator.	PASS
No-show rate	48.46%	$2,423 / 5,000 \times 100 = 48.46\%$.	PASS
Attendance/cancellation counts	Not independently validated	Not stated as a Week 7 validation target in the supplied Week 6 report.	NOT TESTED

VALIDATION CONCLUSION

The headline 48.46% KPI is internally consistent with the documented 2,423 no-shows and 5,000 total appointments. No change to the headline KPI was required.

4. Analytical Validation Results

4.1 Previous No-Show History

The documented no-show rates increase across the main interpretable groups: 43.51% for patients with zero previous no-shows, approximately 53.49% for one, approximately 59.36% for two, and 67.95% for three previous no-shows.

Test	Expected result	Observed/documentated result	Status
Directional pattern	Rates should increase as prior no-show history increases.	$43.51\% \rightarrow \sim 53.49\% \rightarrow \sim 59.36\% \rightarrow 67.95\%$.	PASS
Interpretation	Pattern may support risk stratification, but should not be treated as causal.	Week 6 correctly describes it as an observed association.	PASS
Small groups	Very small categories should not drive the main conclusion.	4 and 5 previous no-show groups were explicitly treated cautiously.	PASS

4.2 Booking Lead Time

The documented rates increase steadily across the five lead-time bands: 24.18%, 30.05%, 33.55%, 43.21% and 60.49%. The difference between the shortest and longest bands is 36.31 percentage points.

Lead-time band	No-show rate	Validation observation
0–2 days	24.18%	Lowest documented rate
3–7 days	30.05%	Higher than 0–2 days
8–14 days	33.55%	Higher than 3–7 days
15–30 days	43.21%	Higher than 8–14 days
31+ days	60.49%	Highest documented rate

VALIDATION CONCLUSION

The lead-time relationship is internally consistent and supports retaining booking lead time as a candidate risk-stratification variable. It does not, by itself, establish that longer lead time causes non-attendance.

4.3 Reminder Channel

The documented no-show rates are 51.39% for no reminder, 49.77% for WhatsApp, 48.41% for email and 45.75% for SMS.

Reminder channel	No-show rate	Week 7 interpretation
None	51.39%	Highest observed rate; descriptive only.
WhatsApp	49.77%	Intermediate observed rate.
Email	48.41%	Intermediate observed rate.
SMS	45.75%	Lowest observed rate; candidate for further testing, not proof of effectiveness.

REFINEMENT MADE

The interpretation is deliberately framed as an observational finding and testing opportunity. The report does not claim that SMS caused the lower no-show rate because reminder assignment may be related to other patient or operational factors.

4.4 Appointment Type

The four appointment-type no-show rates were relatively close in the Week 6 analysis. The Week 7 interpretation therefore remains that appointment type alone should not be used as the primary targeting variable.

5. Dashboard Testing & Refinement

The supplied Week 6 dashboard already had the required decision-oriented structure. For Week 7, the Week 6 page was duplicated so the original remained as the baseline, and a dedicated “Week 7 Testing & Refinement” page was refined. The refinements included overall-rate benchmark lines on Appointment Type and Booking Lead Time, an upward trend line and clearer labels on Previous No-Shows, and revised observed-rate wording and interpretation for Reminder Channel. A Week 7 validation header/panel was also added.

Dashboard element	Test	Result / refinement
KPI card	Does the headline match the documented calculation?	PASS - agrees with the supplied Week 6 report and dashboard evidence.
Appointment Type visual	Does the visual avoid implying a strong difference where rates are close?	PASS - Week 7 adds the 48.46% overall benchmark to improve comparison while retaining the contextual interpretation.
Previous No-Shows visual	Does the visual communicate the increasing pattern?	PASS - Week 7 adds clearer labels and an upward trend line to make the documented progression more visible.
Booking Lead Time visual	Does the visual clearly show the strongest operational gradient?	PASS - Week 7 adds the 48.46% overall benchmark to highlight the 31+ day rate of 60.49%.
Reminder Channel visual	Does wording avoid causal claims?	REFINED - renamed to “Observed No-Show Rates” and revised interpretation to avoid causal claims.
Key Takeaway	Does it identify the strongest observed indicators?	PASS - consistent with the Week 6 Executive Takeaway.
Business Insight/Action panels	Are recommendations connected to evidence?	PASS - consistent with the Week 6 dashboard action panels.

Evidence to attach: the final Week 7 dashboard screenshot/export, with the Week 6 page retained in the PBIX as the baseline. The PBIX remains the primary supporting file.

6. Potential Data Science Contribution & Analytical Handoff

IMPORTANT STATUS

No Data Science collaborator or model output was received during Week 7. The following is a proposed analytical hand-off, not a claim of completed collaboration or model validation.

The Data Analytics findings provide several concrete questions that Data Science could test in a future modelling stage:

Analytics finding	Potential Data Science contribution	What would be gained
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Previous no-shows show a strong observed gradient.	Test previous no-shows as a predictive feature alongside other variables.	Evidence on whether prior behaviour improves individual risk prediction.
Booking lead time rises from 24.18% at 0–2 days to 60.49% at 31+ days.	Test lead time as a feature and investigate non-linear effects or thresholds.	Evidence on whether lead time improves prediction and whether a practical threshold exists.
Reminder channel differs observationally, with SMS at 45.75%.	Evaluate reminder channel while controlling for other characteristics, or test prospectively where appropriate.	Better separation of channel association from possible confounding.
Appointment type rates are relatively close.	Test whether appointment type adds predictive value after behavioural/scheduling variables are included.	Evidence on incremental information value.
Potential interactions have not been assessed.	Test interactions such as prior no-shows × lead time and lead time × reminder channel.	More nuanced risk segmentation.
Current analysis is descriptive.	Evaluate classification performance, calibration and error patterns using an appropriate validation design.	A measurable assessment of predictive usefulness.

Recommended Data Science hand-off: start with previous no-shows and booking lead time as candidate predictors, retain reminder channel as a variable requiring careful evaluation, and compare model results against the descriptive patterns already established in the dashboard.

7. Testing, Refinement & Validation Record

Component	Scenario	Expected	Actual/documentated result	Status	Action / retest
Overall KPI	Reconcile 2,423 no-shows against 5,000.	48.46%.	48.46%.	PASS	No KPI change.
Previous no-shows	Check direction across 0–3 prior no-shows.	Increasing rate.	43.51% → ~53.49% → ~59.36% → 67.95%.	PASS	Retained.
Lead time	Check direction across five bands.	Increasing rate.	24.18% → 30.05% → 33.55% → 43.21% → 60.49%.	PASS	Retained.
Reminder channel	Check ordering and interpretation.	Rates may differ; no causal claim.	51.39% → 49.77% → 48.41% → 45.75%.	PASS	Wording refined.
Appointment type	Check whether differences justify targeting.	Rates relatively close.	Week 6 finding remained stable.	PASS	Contextual visual.
Dashboard usability	Check hierarchy, consistency and action linkage.	Clear decision-support page.	Four equal visuals, KPI, takeaway and panels retained.	PASS	No further structural change required.

8. Limitations & Remaining Analytical Issues

- The underlying dataset is a fixed 5,000-appointment population; stability on new data has not been demonstrated.
- Observed associations do not establish causation.
- Reminder-channel comparisons may be affected by patient characteristics, operational targeting or other confounders.
- Potential interactions between variables have not yet been fully tested.
- Small prior-no-show groups require caution and should not drive the main conclusion.
- No completed Data Science model output was available for comparison.
- A prospective intervention test has not been performed.

9. Week 8 Recommendations

1. Carry the validated 48.46% KPI and Week 7-tested dashboard into final integration.
2. Use previous no-shows and booking lead time as priority candidate variables for predictive validation.
3. Request comparison of descriptive findings with any Data Science model results when available.
4. Investigate whether reminder-channel effects remain after adjustment for relevant characteristics.
5. Use model performance and calibration-not descriptive rates alone-to determine whether individual risk scoring is appropriate.
6. Retain explicit limitations in the final presentation so observed associations are not presented as causal effects.

10. HC-POD Cross-Track Testing Status

MANDATORY CROSS-TRACK REQUIREMENT - NOT YET SATISFIED: The Week 7 assignment requires a meaningful cross-track Test → Finding → Action → Retest → Validated Improvement. The supplied Week 6 evidence documents attempted coordination but no collaborator output. Communication alone is explicitly insufficient, so this report does not claim completion.

TRANSPARENT STATUS

A meaningful cross-track testing activity could not be completed because no collaborator/output was received during the Week 7 submission period. No joint test, model result, exchanged dataset or validated cross-track change is claimed.

The Data Analytics contribution that can be carried forward is an integration-ready hand-off: the dashboard identifies candidate variables, documents the observed patterns, states the limitations and defines the questions that Data Science should test.

Required cross-track item	Week 7 status
Track collaborated with	Data Science - intended dependency; no completed collaboration.
Dependency	Validate whether descriptive no-show patterns improve predictive risk modelling.
Output received	None.
Output provided	Validated analytical findings, candidate variables and testing questions documented in this report.
Testing completed collaboratively	None - not claimed.

Issue/finding	Cross-track dependency remains unresolved.
Action	Prepare explicit Data Science hand-off and validation requirements.
Retest result	Pending collaborator/model availability.
Evidence	Week 7 report, Week 6 dashboard/.pbix and documented communication attempts where applicable.

Requirement	Evidence now	Status	Outstanding
KPI validation	Week 6 report + dashboard	PASS	Raw CSV reconciliation if available.
Dashboard consistency	PBIX structure + screenshot	PARTIAL	Live Power BI filter/calculation execution.
Analytical findings	Week 6 report + visuals	PASS (documented)	Stability on additional data.
Refinement	Interpretation/causal wording refined	PASS	Final Week 7 screenshot captured as the after-state; the original Week 6 page remains the baseline in the PBIX.
Retesting	Documented re-checks of Week 6 outputs	PARTIAL	Retest the refined visuals and wording; underlying KPI/calculations were not changed.
Cross-track testing	Week 6 coordination evidence only	NOT COMPLETED	Actual Test → Finding → Action → Retest with another track.

The Week 7 Data Analytics work is aligned with the official testing/refinement structure, but the evidence set shows two outstanding items: raw-dataset validation where the CSV is unavailable, and the mandatory meaningful cross-track test. These are documented as gaps rather than being represented as completed work.

11A. Evidence-Based Readiness Assessment

12. Conclusion

Week 7 review confirms that the core Week 6 KPI is consistent with the supplied Week 6 report and dashboard evidence and that the principal descriptive patterns remain coherent: previous no-show history and booking lead time show the clearest observed associations with non-attendance. The dashboard interpretation was refined to preserve these findings while avoiding causal claims, particularly for reminder channels.

The principal remaining dependency is predictive and cross-track validation. Data Science could add value by testing whether previous no-shows and booking lead time provide incremental predictive information, examining interactions and confounding, and evaluating whether the observed patterns translate into a reliable risk model. Because no collaborator output was received, these activities are documented as the next-stage hand-off rather than represented as completed work.